

MGT 816 · The Private Firm CFO

So1 — The Private-Company CFO & the Financial Infrastructure
— Exercise Solutions

Posted after class · all three sections

Wed September 2, 2026 · Yale SOM · Prof. John M. Barrios

Exercise: SixWings — how long can we fly? [Section 1]

Task: Solo or with your neighbor, on the half-sheet, find: (a) gross burn, (b) net burn, (c) runway in months + the month we run out of cash, (d) which buys more runway — Option 1 (recurring cut) or Option 2 (one-time collection)? (e) stretch: if we raise in month 6, how big a round buys 18 months of runway at current net burn?



7 minutes — on-screen countdown 7:00

Deliverable: numbers on the half-sheet; be ready to be cold-called.

Section 1 (1:00 · white half-sheets): Cash \$1,200K · payroll \$150K/mo · rent+cloud \$45K/mo · other \$30K/mo · collections \$75K/mo · Option 1: cut \$30K/mo (freeze two open roles) · Option 2: collect \$150K past-due AR today.

'Today' = Sep 1, 2026 cash balance.

Debrief: SixWings runway, worked [Section 1]

Facts

- Cash \$1,200K
- payroll \$150K
- rent+cloud \$45K
- other \$30K
- collections \$75K
- Opt 1: cut \$30K/mo
- Opt 2: collect \$150K today

Gross burn = $150 + 45 + 30 = \$225\text{K}$

Net burn = $225 - 75 = \$150\text{K}$

Runway = $1,200 \div 150 = 8.0 \text{ mo} \rightarrow \text{out of cash} \sim \text{May 1, 2027}$

Option 1: $150 - 30 = 120 \rightarrow 1,200 \div 120 = 10.0 \text{ mo (+2.0)}$

Option 2: $1,200 + 150 = 1,350 \rightarrow 1,350 \div 150 = 9.0 \text{ mo (+1.0)}$

Stretch: cash at mo 6 = $1,200 - 6 \times 150 = 300$; $18 \times 150 = 2,700 \rightarrow$
raise \$2.4M

Runway 8.0 mo (~May 2027) · Option 1 wins (+2.0 vs +1.0) · raise \$2.4M

One-time cash = +1 month, always (it equals one month of net burn). Recurring cuts compound — runway is hyperbolic in burn, only linear in cash.

Exercise: SixWings — how long can we fly? [Section 2]

Task: Solo or with your neighbor, on the half-sheet, find: (a) gross burn, (b) net burn, (c) runway in months + the month we run out of cash, (d) which buys more runway — Option 1 (recurring cut) or Option 2 (one-time collection)? (e) stretch: if we raise in month 6, how big a round buys 18 months of runway at current net burn?



7 minutes — on-screen countdown 7:00

Deliverable: numbers on the half-sheet; be ready to be cold-called.

Section 2 (2:35 · blue half-sheets): Cash \$1,800K · payroll \$200K/mo · rent+cloud \$60K/mo · other \$40K/mo · collections \$100K/mo · Option 1: cut \$40K/mo · Option 2: collect \$200K AR today.

'Today' = Sep 1, 2026 cash balance.

Debrief: SixWings runway, worked [Section 2]

Facts

- Cash \$1,800K
- payroll \$200K
- rent+cloud \$60K
- other \$40K
- collections \$100K
- Opt 1: cut \$40K/mo
- Opt 2: collect \$200K today

Gross burn = $200 + 60 + 40 = \$300K$

Net burn = $300 - 100 = \$200K$

Runway = $1,800 \div 200 = 9.0 \text{ mo} \rightarrow \text{out of cash } \sim \text{Jun 1, 2027}$

Option 1: $200 - 40 = 160 \rightarrow 1,800 \div 160 = 11.25 \text{ mo (+2.25)}$

Option 2: $1,800 + 200 = 2,000 \rightarrow 2,000 \div 200 = 10.0 \text{ mo (+1.0)}$

Stretch: cash at mo 6 = $1,800 - 6 \times 200 = 600$; $18 \times 200 = 3,600 \rightarrow \text{raise } \$3.0M$

Runway 9.0 mo (~Jun 2027) · Option 1 wins (+2.25 vs +1.0) · raise \$3.0M

One-time cash = +1 month, always (it equals one month of net burn). Recurring cuts compound — runway is hyperbolic in burn, only linear in cash.

Exercise: SixWings — how long can we fly? [Section 3]

Task: Solo or with your neighbor, on the half-sheet, find: (a) gross burn, (b) net burn, (c) runway in months + the month we run out of cash, (d) which buys more runway — Option 1 (recurring cut) or Option 2 (one-time collection)? (e) stretch: if we raise in month 6, how big a round buys 18 months of runway at current net burn?



7 minutes — on-screen countdown 7:00

Deliverable: numbers on the half-sheet; be ready to be cold-called.

Section 3 (4:10 · yellow half-sheets): Cash \$2,100K · payroll \$240K/mo · rent+cloud \$70K/mo · other \$40K/mo · collections \$140K/mo · Option 1: cut \$35K/mo · Option 2: collect \$210K AR today.

'Today' = Sep 1, 2026 cash balance.

Debrief: SixWings runway, worked [Section 3]

Facts

- Cash \$2,100K
- payroll \$240K
- rent+cloud \$70K
- other \$40K
- collections \$140K
- Opt 1: cut \$35K/mo
- Opt 2: collect \$210K today

Gross burn = $240 + 70 + 40 = \$350K$

Net burn = $350 - 140 = \$210K$

Runway = $2,100 \div 210 = 10.0 \text{ mo} \rightarrow \text{out of cash } \sim \text{Jul 1, 2027}$

Option 1: $210 - 35 = 175 \rightarrow 2,100 \div 175 = 12.0 \text{ mo (+2.0)}$

Option 2: $2,100 + 210 = 2,310 \rightarrow 2,310 \div 210 = 11.0 \text{ mo (+1.0)}$

Stretch: cash at mo 6 = $2,100 - 6 \times 210 = 840$; $18 \times 210 = 3,780 \rightarrow$
raise \$2.94M

Runway 10.0 mo (~Jul 2027) · Option 1 wins (+2.0 vs +1.0) · raise \$2.94M

One-time cash = +1 month, always (it equals one month of net burn). Recurring cuts compound — runway is hyperbolic in burn, only linear in cash.